GGG-T-001310(GSA-FSS)
December 27, 1967
INTERIM REVISION OF
Fed. Spec. QQ-T-40
August 26, 1953

INTERIM FEDERAL SPECIFICATION

TAPE, FISH, STEEL

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification QQ-T-40, dated August 26, 1953.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers steel fish tapes with reels and accessories.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity which are available, but only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and styles. Fish tape shall be of one type, and of the following styles as specified (see 6.1):

Type I. - Steel, with reel.
  Style 1. - Flat.
  Style 2. - Round.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:
  Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
  Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

FSC 5120

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are for the convenience of identification, and are not intended to exclude the purchase of fish tapes which are otherwise in accordance with this specification.

3.2 Design and construction. The fish tape shall consist essentially of a steel tape enclosed in a metal or plastic reel, as specified (see 6.1).

3.3 Type I, style 1. Style 1 fish tape shall be flat, oil-tempered, spring-steel wire with an edge radius of 0.030 to 0.035 inch. One end of the tape shall be fastened to the inside of the reel in a manner to permit easy replacement. The other end shall be looped, or otherwise designed to accommodate a minimum of five No. 10 AWG bare copper conductors, and shall withstand the test specified in 4.4.2. The overall width of the loop shall be 5/16 inch, plus or minus 1/32 inch, to permit clearance in negotiating bends in conduit. When specified (see 6.1), tape shall be furnished with a guide ball instead of a loop. Tape shall be of the dimensions specified (see table I and 6.1).

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TABLE I. Type I, styles 1 and 2 fish tapes

Tape size designation	Type	Tape dimensions (inches)			Tape length (feet)	Tensile strength p.s.i.	Elongation in 10 inches (max. percent)
		Width ± 0.005	Thickness ± 0.003	Diameter ± 0.002	± 6 in.		
1	Flat	1/8	0.047		50	230,000 - 280,000	5
2	"	1/8	.062		50	"	5
3	"	1/8	.047		100	"	5
4	"	1/8	.062		100	"	5
5	"	1/8	.062		200	"	5
6	"	1/4	.062		100	"	5
7	"	1/4	.062		200	"	5
8	Round			0.080	50	280,000 - 300,000	5
9	"			.080	100	"	5
10	"			.080	200	"	5

3.4 Type I, style 2. Style 2 fish tape shall be round, oil-tempered, spring-steel wire with a guide ball on the outer end. The opposite end of the tape shall be fastened to the inside of the reel in a manner to permit easy replacement. Dimensions shall be as specified in table I and 6.1. When specified (see 6.1), outer end of tape shall be provided with a loop instead of a guide ball.

3.5 Reel. A reel shall be provided to completely contain the coiled fish tape (except for looped end or guide ball), and prevent it from unwinding when not in use. The reel shall be constructed of steel or plastic as specified (see 3.5.1, 3.5.2, and 6.1). When assembled, the reel shall form a hollow ring with a continuous slot around the outer periphery as illustrated in figure 1. A hook, slot, or other means of fastening, shall be provided in the inner diameter of the reel to permit engagement or disengagement of the tape for replacement purposes. The outer diameter of the reel shall be not less than 7 inches nor more than 14 inches. The complete assembly shall be tested for compliance with 4.4.

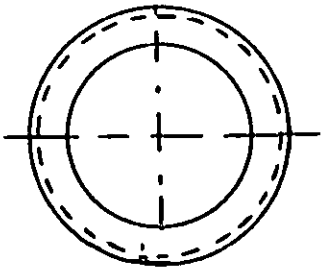


FIGURE 1. Reel.

3.5.1 Steel reel. Steel reel shall be constructed of two drawn steel parts having a material thickness of 0.022 to 0.037 inch. The two parts shall be held together by welding or crimping. The slot around the outer periphery shall permit smooth and easy winding and unwinding of tape, but shall maintain tension on the tape to prevent it from springing loose. When specified (see 6.1), a winder shall be furnished with the reel. Winder may be of metal, or plastic having the same high-impact resistance as the plastic reel (see 3.5.2). It shall slide smoothly and easily around the reel.

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3.5.2 Plastic reel. Plastic reel shall be furnished with a plastic winder. Both shall be of high quality that will meet the impact strength test specified in 4.4.4. The reel shall have a side wall minimum thickness of 0.090 inch for reels retaining 100 feet or more of tape, and a minimum of 0.085 inch for 50-foot lengths of tape. The reel shall provide a satisfactory case for the tape. A slot around the outer periphery of the reel, together with the winder, shall maintain tension on the tape and retain it in the reel. The winder shall be of the replaceable type, and slide smoothly and easily around the reel for winding the tape.

3.6 Guide ball. The guide ball shall be of steel designed to provide smooth passage over obstructions in the conduit. Each guide ball shall consist of a threaded male segment and a threaded female segment similar to figure 2. The eye of the guide ball shall be of sufficient size to accommodate a minimum of five No. 10 AWG bare copper conductors. The opposite end of the guide ball for flat tape shall have an opening large enough to swivel on 1/8 by 0.062-inch maximum tape. When a guide ball is specified for 1/4-inch tape (see 6.1), the opening in the guide ball shall be of proper size to permit the wire to swivel. The opposite end of the guide ball for round tape shall have an opening large enough to swivel on 0.082-inch maximum diameter tape. The guide ball shall withstand the test specified in 4.4.3.

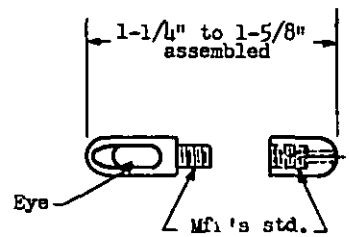


FIGURE 2. Guide ball.

3.7 Replacement tape, without reel. Replacement tape, without reel, shall be furnished as specified in table I and 6.1, and shall be in accordance with the requirements of 3.3 or 3.4, as applicable.

3.8 Finish and coating.

3.8.1 Tapes. Tapes shall be rustproofed and free from nicks, kinks, bends or any defect which may impair serviceability.

3.8.2 Reels and guide balls. Reels and guide balls shall be free from nicks, burrs, cracks, or other imperfections, and all metal parts shall be painted, plated, or otherwise treated to resist corrosion.

3.9 Identification marking. The reels and guide balls shall be marked in a legible and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.10 Workmanship. Tapes, reels, and guide balls shall be free from any defects or imperfections that might affect serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

TABLE II. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	4.0	Defects per 100 units	4.3.1
Dimensional examination	1 each	S-3	1.5	Defects per 100 units	4.3.2
Testing (see 4.4).					
Preparation for delivery	One container	S-2	4.0	Defects per 100 units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. One sample unit shall be tested in accordance with 4.4.1 through 4.4.4.

4.4.1 Tensile and elongation tests for fish tapes. One sample unit shall be tested for tensile strength and elongation. A 6-foot maximum length of tape should be adequate for inspecting and testing in accordance with method 211.1 of Fed. Test Method Std. No. 151, and table I.

4.4.2 Loop and reel pull test. A minimum of five No. 10 AWG bare copper conductors (of sufficient length for gripping and performing the pull test) shall be inserted in the looped end of the tape, twisted around themselves and then securely wrapped with electricians' tape. A minimum of eight loops of fish tape shall snub upon the inner spool and upon themselves. The sample reel shall be clamped in a manner to hold it firmly, and a 200-pound pull applied to the conductors. Any deformation of the tape or reel shall be considered cause for rejection.

4.4.3 Tape and guide ball assembly test. The assembled tape and guide ball shall be subjected to a test in accordance with standard laboratory practice to determine the capability of the assembly to remain intact when a 400-pound pull is applied.

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4.4.4 Plastic test. Plastic materials used for reels and winders must meet the following requirements when tested in accordance with ASTM D 256-56, method A:

Impact strength, minimum foot-pounds per inch of notch at 23° C. $\pm$ 2° C. (73.4° $\pm$ 3.6° F.) shall be 1.0.

4.5 Inspection of preservation. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size; location; sequence; or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), fish tapes shall be preserved, packaged, and packed in accordance with MIL-H-15424 (as modified by 5.1.1). The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.1).

5.1.1 As part of the requirements of section 5 of this specification, table IV is considered as part of table I of MIL-H-15424.

TABLE IV

Col. 1	Col. 2	Col. 3	Col. 4	Col. 8
Tape, fish and accessories	P-8 P-9	I	1	Quantity of items packed per shipping container shall be such as to meet the weight requirements of 3.4.1 (MIL-H-15424)

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Style required (see 1.2.1).
(c) If guide ball is required for flat tape with reel (see 3.3).
(d) Size tapes required (see table I).
(e) If loop is required instead of guide ball for round tapes with reel (see 3.4).
(f) If winder is desired for reel (see 3.5).
(g) Specify if guide ball is required for 1/4-inch tape (see 3.6).
(h) Specify type of replacement tape required (see 3.7).
(i) Selection of applicable level of preservation, packaging, and packing (see 5.1).

6.2 Supersession data. The classification of tapes covered in this specification corresponding to those covered in QQ-T-40 are as follows:

QQ-T-40	GGG-T-001310(GSA-FSS)
Form I. - Flat fish tape, reel and puller combination.	Type I. - Steel, with reel. Style 1. - Flat.
Form II. - Round fish tape with guide balls, reel and puller combination.	Style 2. - Round.
Form III. - Coil-spring-type fish tape.	Deleted.
Class 1. - Complete fish tape.	Deleted.
Class 2. - Fish tape leader.	Deleted.
Form IV. - Accessories.	Included in section 3 requirements of specification.
Class 1. - Guide balls, for flat tape.	
Class 2. - Guide balls, for round tape.	
Class 3. - Tapes without reels or for replacement.	

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